

Cybersecurity, IT Act & Security Controls

Lesson 56 — Law, Threats and Controls

JKSSB Computer Awareness

October 8, 2026

Why Security Matters: The CIA Triad

- ▶ A computer holds data worth protecting: marks, accounts, personal records, and official files.
- ▶ **Security** means protecting data and systems against theft, damage, and misuse.
- ▶ Three goals, remembered as **CIA**: **confidentiality** (only authorised persons read), **integrity** (data is not altered), **availability** (systems stay usable).
- ▶ Every control in this lesson serves one or more of these goals.

The one idea

Threats attack the CIA triad; controls defend it. Match each control to what it does.

The IT Act, 2000

- ▶ The **Information Technology Act, 2000** is India's law dealing with electronic records, electronic governance, and cybercrime.
- ▶ It gives **legal validity** to electronic records and electronic signatures, so online transactions and e-governance have a legal basis.
- ▶ It defines **cyber contraventions and offences**, such as unauthorised access, damage to computer systems, and fraud committed through a computer.
- ▶ Year and scope are the exam facts: the Act is of **2000** and covers **electronic records, e-governance, and cybercrime**.

Trap alert

The IT Act is of **2000**, not 2008 as a new Act — later changes were amendments to the 2000 Act. Do not invent section numbers.

The Malware Umbrella

- ▶ **Malware** (malicious software) is the umbrella term for any program written to harm a computer, steal data, or disrupt work.
- ▶ The four named members: **virus**, **worm**, **Trojan horse**, and **ransomware**.
- ▶ Related nuisance programs: **spyware** (secretly watches and reports activity) and **adware** (shows unwanted advertisements).
- ▶ Exam wording: malware is the **general** term; virus, worm, Trojan, and ransomware are **specific** types under it.

Keep the pair straight

Malware is the umbrella; each named threat is one rib of the umbrella.

Virus: Needs a Host and Human Action

- ▶ A **virus** attaches itself to a host file or program and spreads when that file is run or shared.
- ▶ It **needs human action** — opening an infected attachment or running an infected file — to spread.
- ▶ It can corrupt, delete, or alter files once it runs.
- ▶ Office-desktop picture: an infected document copy passed on a pen drive infects the next machine that opens it.

Exam contrast

Virus needs a **host file plus human action**; worm does not.

Worm: Self-Spreading Across Networks

- ▶ A **worm** is standalone malware that **copies itself** and spreads across a network **without** a host file.
- ▶ It needs **no human action** to move from machine to machine; it exploits the network itself.
- ▶ Worms consume **bandwidth and system resources**, slowing or crashing networks.
- ▶ One infected machine can flood the office network while nobody opens anything.

Virus vs worm

Virus = host file + human action. Worm = self-copying, no host, no click needed.

Trojan Horse: Looks Useful, Hides Harm

- ▶ A **Trojan horse** pretends to be useful software but hides a harmful purpose inside.
- ▶ Unlike a virus or worm, it **does not replicate** by itself.
- ▶ The harm: it opens a **backdoor**, letting an attacker steal passwords, spy on activity, or control the machine.
- ▶ Office-desktop picture: a “free” utility that installs fine — and quietly hands the attacker the keyboard.

Trap alert

Trojan does **not** self-replicate. Replication belongs to virus and worm.

Ransomware: Locks, Then Demands Payment

- ▶ **Ransomware encrypts** the victim's files or locks the screen, then demands payment (**ransom**) for release.
- ▶ Payment does **not** guarantee recovery; the defence is a clean **backup** kept separate from the machine.
- ▶ It commonly arrives through phishing mail and infected downloads.
- ▶ Exam one-liner: ransomware = locks data + demands money.

Worked contrast

Virus corrupts files; worm floods the network; Trojan spies through a backdoor; ransomware locks files and demands payment.

Antivirus and Quarantine

- ▶ **Antivirus** software detects, blocks, and removes malware, using known malware signatures and suspicious-behaviour checks.
- ▶ It must be **updated regularly**, since new malware appears every day.
- ▶ **Quarantine** means isolating a suspicious file so it cannot run or spread while awaiting a decision.
- ▶ Quarantine is **isolation**, not deletion and not a backup.

Keep the pair straight

Delete removes; quarantine isolates. An isolated file can still be inspected or restored.

Phishing and Anti-Phishing

- ▶ **Phishing** is a fake message (mail, SMS, or site) that impersonates a trusted sender to steal passwords or money.
- ▶ Warning signs: urgent threats, unexpected attachments, links that do not match the claimed sender, requests for OTPs or passwords.
- ▶ **Anti-phishing** defences: verify the sender, hover over links before clicking, never share OTPs, and report the message.
- ▶ A genuine bank or office **never** asks for a password or OTP by mail or message.

Trap alert

Phishing steals **credentials by deception**; it is not a virus and needs no malware to succeed.

Encryption vs Digital Signature

- ▶ **Encryption** converts readable data (plaintext) into coded form (ciphertext) so only authorised persons can read it — it gives **confidentiality**.
- ▶ A **digital signature** proves who sent a message and that it was not altered — it gives **authenticity and integrity**.
- ▶ A signature does **not** hide the message; encryption does.
- ▶ This repeats the Lesson-16 email rule (TRAP-15): signature $\neq$ encryption.

One line each

Encryption hides content. Signature proves sender and intactness.

SSL and TLS: Secure Transmission

- ▶ **SSL** stands for Secure Sockets Layer; **TLS** stands for Transport Layer Security.
- ▶ TLS is the **successor** of SSL; the pair provides a secure, encrypted channel over a network.
- ▶ They protect data **in transit**: mail sending, web login, and online forms (the padlock in the browser address bar).
- ▶ Exam anchor (WO 2026 Q76): the secure email transmission protocol is **SSL/TLS**.

Trap alert

SSL/TLS protect data in transit. They do not clean an infected file and do not replace antivirus.

Firewall: The Network Barrier

- ▶ A **firewall** is a barrier that filters incoming and outgoing network traffic against a set of rules.
- ▶ It **blocks unauthorised access** while letting permitted traffic pass.
- ▶ It can be **hardware**, **software**, or both.
- ▶ Picture it as the gatekeeper at the office network door, checking every packet against the list.

Worked contrast

Firewall filters **traffic** at the boundary; antivirus cleans **malware** inside the machine. They complement, not replace, each other.

Patching and MFA: Two More Controls

- ▶ **Patching** means installing vendor-supplied software updates that fix security flaws; an unpatched machine stays open to known attacks.
- ▶ **MFA** stands for Multi-Factor Authentication: proving identity with **two or more** checks, e.g. password plus OTP or fingerprint.
- ▶ MFA protects **logins**; patching protects **software flaws**.
- ▶ Strong, distinct **passwords** remain the first factor — MFA adds factors, it does not replace the password.

Which control does what

Patching closes holes in software. MFA closes the door against stolen passwords.

Which Control Does What: Matching Map

Control	Job
Firewall	Filters network traffic; blocks unauthorised access
Antivirus	Detects and removes malware on the machine
Quarantine	Isolates a suspicious file so it cannot run
Encryption (SSL/TLS)	Protects data in transit from reading
Digital signature	Proves sender and intactness
Patching	Fixes security flaws in software
MFA	Adds extra login checks beyond the password
Backup	Recovers data after ransomware or loss

Read the stem's verb — filters, cleans, isolates, hides, proves, fixes, or recovers — then pick the matching row.

Trap Alert: Security Facts to Keep Straight

- ▶ IT Act = **2000**; covers electronic records, e-governance, and cybercrime.
- ▶ Malware is the **umbrella**; virus, worm, Trojan, and ransomware are types under it.
- ▶ Virus needs host + action; worm self-spreads; Trojan does **not** replicate; ransomware locks and demands payment.
- ▶ Digital signature $\neq$ encryption: signature proves, encryption hides. (TRAP-15)
- ▶ SSL/TLS = secure **transmission**; firewall = traffic **barrier**; quarantine = **isolation**; MFA = **extra login checks**.

Lesson 56: Recall

- ▶ The Information Technology Act, 2000 gives legal validity to electronic records and signatures and defines cyber contraventions and offences.
- ▶ Malware is the umbrella term; virus (host + action), worm (self-spreading), Trojan (false-useful backdoor, no replication), and ransomware (locks + demands payment) sit under it.
- ▶ Antivirus detects and removes malware; quarantine isolates a suspicious file; patching fixes software flaws.
- ▶ Phishing steals credentials by deception; verify senders, never share OTPs.
- ▶ Encryption gives confidentiality; digital signature gives authenticity and integrity; SSL/TLS secure data in transit; firewall filters traffic; MFA adds login factors.